

SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering



ELEC 490/498 Project Proposal

Computer-Vision Autonomous Golf Cart

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Submitted by Group 07

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Executive Summary

[Paragraph 1: Identify the problem, intended users or market, proposed system, and expected benefit. State the project goal and the end product.]

[Paragraph 2: Summarize the principal functions and performance targets, the proposed design and production approach, and two or three milestones.]

[Paragraph 3: List the prototypes. Summarize the testing approach, resources and total estimated cost in the stated currency, and readiness to begin the project. If further research and / or learning is necessary, list the topics and their applicability.]

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Acronyms

ASCC Autonomous Smart Caddy Cart

GPS Global Positioning System

1. Introduction

1.1 The shortfalls of autonomous golf carts

Walking golfers can already buy electric push carts that follow or lead the way on the course [1]. So, the problem of carrying your own clubs has already been solved, but where these carts fall short is the ability to actually guide how a golfer plays the game. The carts can't act like a human caddy would, which is where the purpose of the Autonomous Smart Caddy Cart (ASCC) comes in. Most golfers either track their information on paper, get info from their watch, phone, or a range finder [2], [3], [4]. The ASCC is an all in one solution to both club carrying, and caddy assistance. A golfer would usually need to manually input, or accept their distance, club information on their phone or watch. The ideal user of this cart is a golfer who wants to track their information, stay off their phone, and have a cheaper solution than the cost of a human caddy and electric cart combined.

1.2 A cost effective, phone-free autonomous smart caddy

The objective of this project is to design and prototype a simple tracking golf caddy that follows the golfer safely, identifies club usage without any input, and gives personalized advice to the golfer through a display and or speaker. Each outcome is measurable. The cart must hold its following distance (PF-03), respond to obstacles (PF-05), and detect tagged clubs (IF-04) within the targets in Table 3, and its shot-distance estimates and missing-club warnings are checked by the tests in Section 4. The engineering involved is motor control, embedded sensing, golfer tracking, Global Positioning System (GPS) processing, and interface integration. Figure 1 shows the system in its operating context.

Person-following robots track their target with cameras, laser rangefinders, or radio ranging, and the choice trades cost against robustness to lighting, occlusion, and bystanders [5]. Commercial following carts avoid the perception problem by tracking a handset that the golfer carries [1]. For shot tracking, Hall et al. tested two club-tag GPS systems against laser and tape measurements and found typical distance errors of three yards or less, but one system missed 67 shots where the other missed 3 [4]. Detecting the shot event is therefore the harder problem, not measuring its length.

The ASCC differs from these systems in where it senses. It detects a club leaving the bag instead of the swing, and it measures from the cart's position instead of the golfer's. This removes the phone and watch, at the cost of two errors that the test plan must quantify:

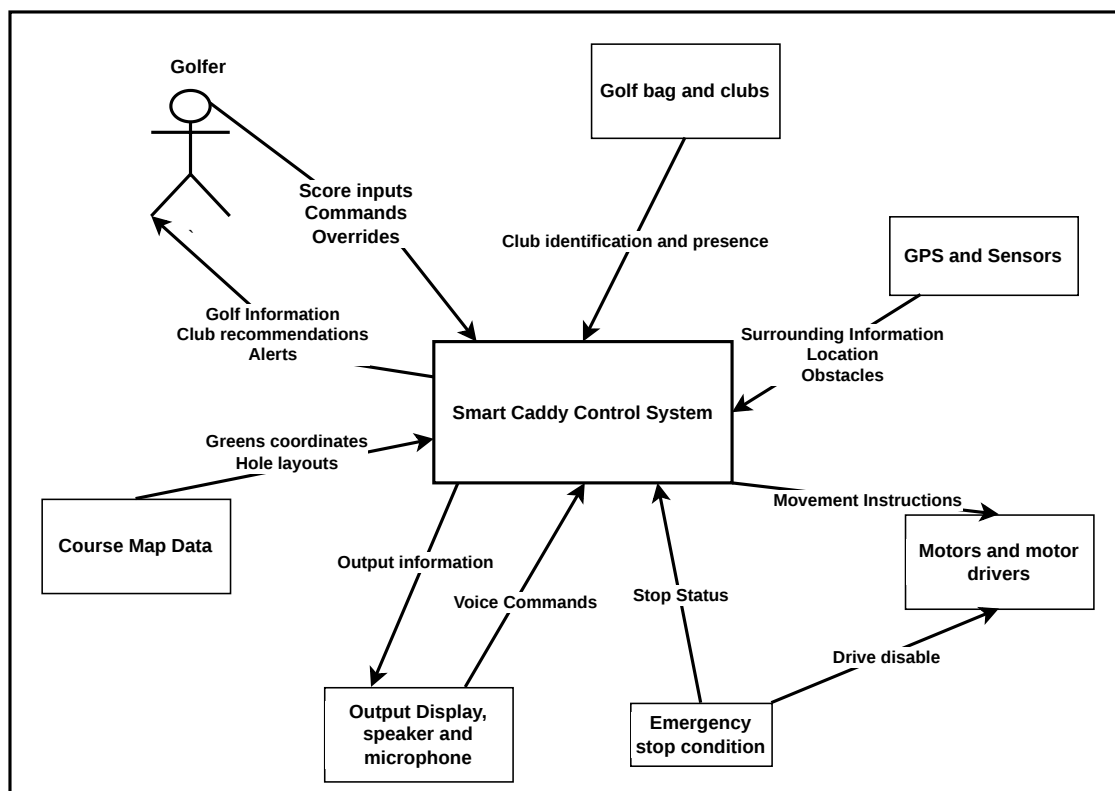


Figure 1: System context diagram for the Smart Caddy control system, showing the external entities it interacts with and the information exchanged with each.

a club removal that is not a shot, and the offset between the cart and the ball. Table 1 compares the three designs feature by feature.

Table 1: Comparison of the proposed ASCC design with existing following-cart and shot-tracking products.

Design consideration	Stewart Golf Q Follow [1]	Arccos Smart Sensors with Apple Watch [2], [3]	Proposed ASCC design
Equipment transport	Motorized cart carries the golf bag	Separate bag-transport solution is required	Motorized cart carries the bag and the golf-assistance hardware
Golfer following	Follows a carried Bluetooth handset; also supports remote control	No powered following hardware	Tracks one golfer and maintains a specified following distance
Club-use detection	None	Club-mounted sensors identify the club used on detected shots	Bag-mounted reader detects tagged-club removal and return
Shot distance and advice	Requires separate equipment	Associates shots with GPS locations; watch recommends clubs	Estimates distances from cart GPS locations at club events; onboard interface recommends clubs

1.3 Deliverables: follow-me mobility, smart club tracking, and onboard caddy assistance

The deliverable is a single working prototype cart, demonstrated outdoors on open, dry grass with one golfer and one bag of tagged clubs. Its inputs are the golfer's position, button commands, club tag readings, GPS fixes, and obstacle detections; its outputs are wheel motion, displayed information, and warnings. The subsystems meet at the club event: a removal triggers a position record, and the distance to the next removal becomes the previous shot's estimated length. Tagging clubs in this way is a strategy also used by commercial shot trackers [2]. Table 2 lists each deliverable with its assumptions and limitations.

Table 2: Project deliverables, operating assumptions, and limitations.

Deliverable	Included functionality	Operating assumptions / constraints	Limitations / exclusions
Following-cart subsystem	Powered transport, one-golfer following, basic obstacle stopping, manual override	Dry grass; defined terrain and speed limits; golfer detectable by the tracking system	No independent course navigation; restricted terrain capability
Smart-bag subsystem	Tagged-club identification, removal/return events, missing-club warnings	One bag of tagged clubs; reliable reads within the defined bag configuration	Club removal does not confirm a shot; multiple-club handling requires evaluation
Golf-assistance subsystem	Estimated shot distances, player-specific club history, club recommendations	Usable GPS; identifiable successive shot events; green coordinates loaded beforehand	GPS error and cart-to-ball offset affect distance estimates; reliable putt measurement excluded
Onboard interface	Display and physical controls for core tasks	Information and controls accessible without phone interaction during play	Voice interaction remains an extension unless committed as a requirement
Integrated powered prototype	One working cart demonstrated with all core subsystems	Battery sized for the specified demonstration; dry-weather testing	Waterproofing and full 18-hole endurance excluded

2. Functional Description

2.1 System overview

Show the functional, physical, and logical diagrams.
Label components, inputs, outputs, arrows, and signal types or protocols.

Figure 2: [System architecture and information flow. (This is a generic example. Use several diagrams to show the various relationships between system components.)]

2.2 Interface and performance specifications

The interface and performance specifications defined in Table 3 establish the measurable quantitative targets required to ensure the smart caddy effectively alleviates physical strain and cognitive load for senior golfers. Physical and hardware interfaces focus on ergonomic accessibility, payload capacity, and reliable user tracking (**IF-01** through **IF-04**). Performance metrics guarantee that the mechatronic chassis maintains safe following distances, handles standard 18-hole course topographies, and reliably detects active club removal and return (**PF-01** through **PF-05**). Each specification is tagged with a unique identifier to facilitate downstream verification and validation in Section 4.

Table 3: Preliminary system specifications.

ID	Requirement	Target	Tolerance
IF-01	User Tracking Beacon Range	15.0 m	± 2.0 m
IF-02	Onboard Display Diagonal Size	7.0 in	Minimum 5.0 in
IF-03	Bag Carrier Weight Capacity	20.0 kg	+5.0 / -0.0 kg
IF-04	Club RFID Detection Range	15.0 cm	± 5.0 cm
PF-01	Continuous Operating Range (Single Charge)	12.0 km	± 1.5 km
PF-02	Maximum Autonomous Cruise Speed	5.0 km/h	± 0.5 km/h
PF-03	Target Following Distance	1.5 m	± 0.3 m
PF-04	Maximum Incline In-Motion Gradeability	15.0 deg	Minimum 12.0 deg
PF-05	Obstacle Detection Response Delay	200 ms	≤ 300 ms

2.3 Risks and safety hazards

3. Design and Production Approach

3.1 Retroencabulator development

The autonomous golf pushcart will be developed through system design, subsystem prototyping, integration, testing, and refinement. The team has already obtained some technical components and will attempt to purchase a used motorized golf cart as the base

platform. The Raspberry Pi, sensors, motor-control electronics, and smart-bag system will be integrated onto the cart, with development focused on motor control, sensor-based golfer following, RFID-based club identification and tracking. A controller will be used to process sensor inputs and control the cart's movement during golfer following and obstacle avoidance. Each subsystem will be tested independently before being integrated into the complete system. Testing will begin in a controlled environment before progressing to outdoor conditions involving different walking speeds, obstacles, and terrain, with results used to refine the system and prioritize the MVP requirements.

3.2 Milestones and division of labor

Table 4: Technical milestones and ownership.

ID	Milestone	Due date	Lead
M1	Motorized control of the golf cart	Oct 30th, 2026	Connor
M2	Raspberry Pi integration and hardware intergration of the necessary sensors	Nov 20th, 2026	Vince
M3	RFID-Based Club Identification and Tracking	Dec 18th, 2026	Jp
M4	Obstacle detection/avoidance/emergency stop functionality	Jan 22nd, 2027	Vince
M5	User Interface Design and display	Feb 5th, 2027	Jp
M6	Full Subsystem Integration and System Refinement	Mar 1, 2027	All members
M7	Testing and validation of retroencabulator complete	Mar 15, 2027	All members

3.3 Contingencies

The main project risks include difficulty obtaining a suitable motorized cart, unreliable sensor-based golfer following, limitations in RFID detection, motor issues, and delays during system integration. If a component fails to meet requirements during testing, the team will modify the software or use an alternative sensor, controller, or RFID configuration. If RFID tags cannot be reliably detected within the golf bag, the team will test alternative identification tags such as NFC tags for example. If a suitable motorized golf cart cannot be obtained, a different cart or drive platform will be considered. Outdoor testing may be moved to a controlled environment if weather or terrain prevents reliable testing. If

the project falls behind schedule, the team will prioritize the MVP features of basic cart movement, golfer following, obstacle detection, emergency stopping, and RFID-based club identification and tracking, while postponing more advanced features such as shot-based club recommendations or mobile-app functionality. These alternatives may add some cost or development time but will help ensure that the core system is completed.

4. Testing and Evaluation

Insert diagrams showing the test environments, and link IDs to the requirements and specifications.
Show what will be tested, the main inputs and measurements, and how the results connect to the project requirements.

Figure 3: [Proposed system testing approach...be more specific here.]

Requirement	Proposed test	Expected evidence of success
[ID]	[Briefly describe the planned test.]	[State the expected result or pass criterion.]

5. Resource Requirements

5.1 Scheduling access to resources

Table 5: Resource access by milestone.

Milestone	Resource / source	Access date / lead time
M1	Motors, Controller, Golf Cart	Nov 1st, 2026; ASAP
M2	Raspberry Pi, Ultrasonic/Lidar Sensors,	Nov 1st, 2026; ASAP
M3	RFID readers/tags, User Interface Display	Dec 1st, 2026; 2-3 weeks in advance
M4	Outdoor testing area	Feb 1st, 2027; 1 week in advance

5.2 Materials and other resources

Table 6: Preliminary cost estimate (600 CAD).

Item / source	Qty.	Unit cost	Total
Raspberry Pi 5 - 4G	1	Existing	0 CAD
Ultrasonic Sensors	2	Existing	0 CAD
RFID Read Tags	2	10 CAD (5)	20 CAD
Lidar Sensor	1	50 CAD	50 CAD
Used Motorized Golf Cart	1	250 CAD	250 CAD
Motor controller	1	50 CAD	50 CAD
Battery/Power Supply	1	75 CAD	75 CAD
Mounting Hardware	1	30 CAD	30 CAD
Wiring, Connectors, and Switches	1	30 CAD	30 CAD
Tax and Shipping	-	-	75 CAD
Contingency	-	-	15 CAD
Estimated total			595 CAD

6. Conclusion

[Identify the proposed system and its purpose. Recap the objectives, scope, principal specifications, design strategy, milestones, testing approach, resources, and estimated cost. Close with the expected benefit for the intended user or application.]

7. References

- [1] Stewart Golf. “Q Follow electric golf trolley,” Accessed: Jul. 15, 2025. [Online]. Available: <https://www.stewartgolf.co.uk/products/stewart-golf-q-follow>
- [2] Arccos Golf. “Arccos Smart Sensors,” Accessed: Feb. 17, 2025. [Online]. Available: <https://arccosgolf.com/products/smart-sensors>
- [3] Arccos Golf. “Arccos for Apple Watch,” Accessed: Jul. 27, 2021. [Online]. Available: <https://arccosgolf.com/pages/apple-watch>
- [4] T. E. D. Hall, A. G. B. Willmott, N. J. Hynes, J. E. T. Wells, and P. M. Allen, “The validity and reliability of two golf performance-tracking Global Positioning System (GPS) devices,” *International Journal of Golf Science*, vol. 13, no. 1, pp. 103–121, 2025. [Online]. Available: <https://www.golfsciencejournal.org/article/144587>
- [5] M. J. Islam, J. Hong, and J. Sattar, “Person-following by autonomous robots: A categorical overview,” *The International Journal of Robotics Research*, vol. 38, no. 14, pp. 1581–1618, Dec. 2019. DOI: [10.1177/0278364919881683](https://doi.org/10.1177/0278364919881683)